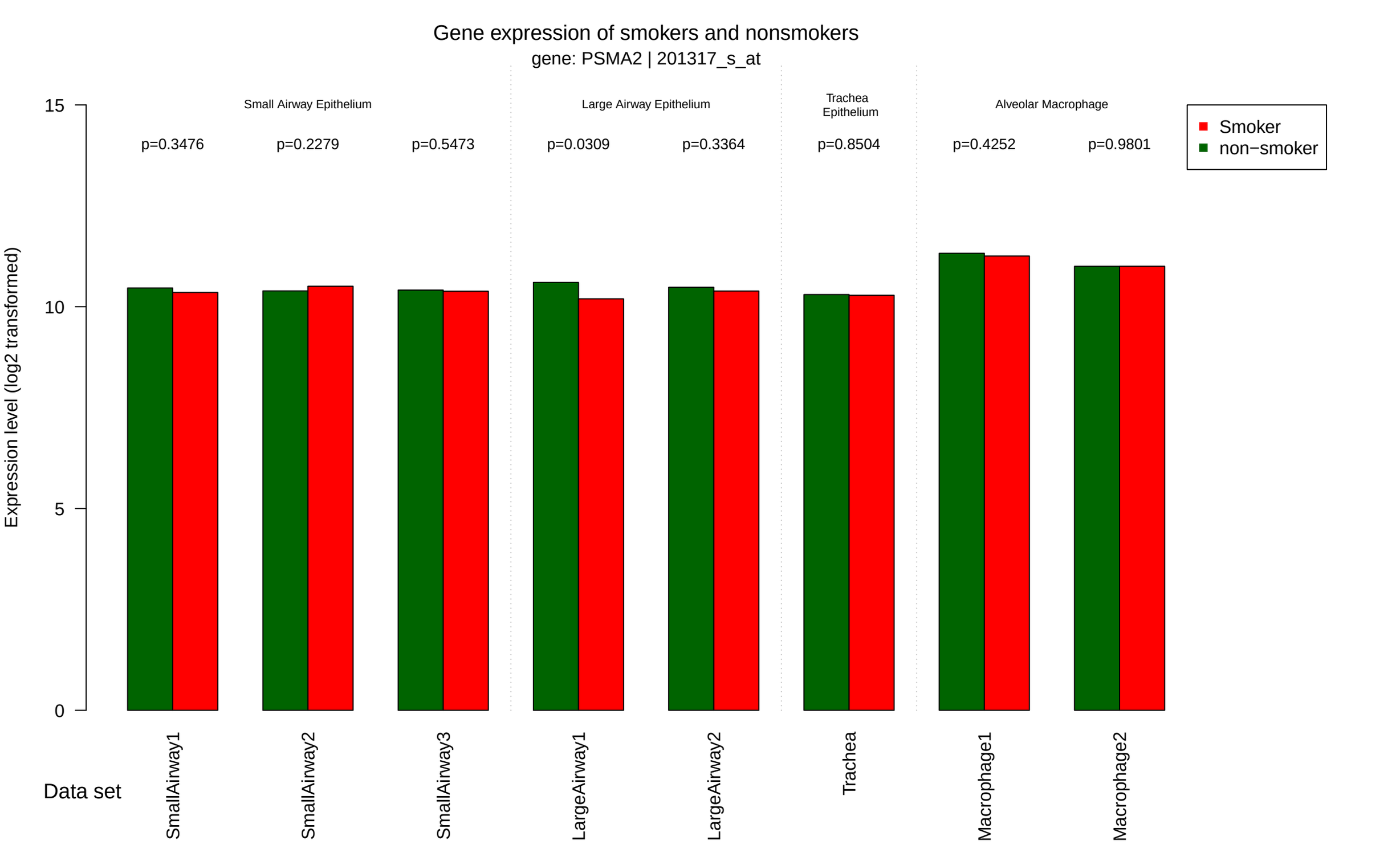
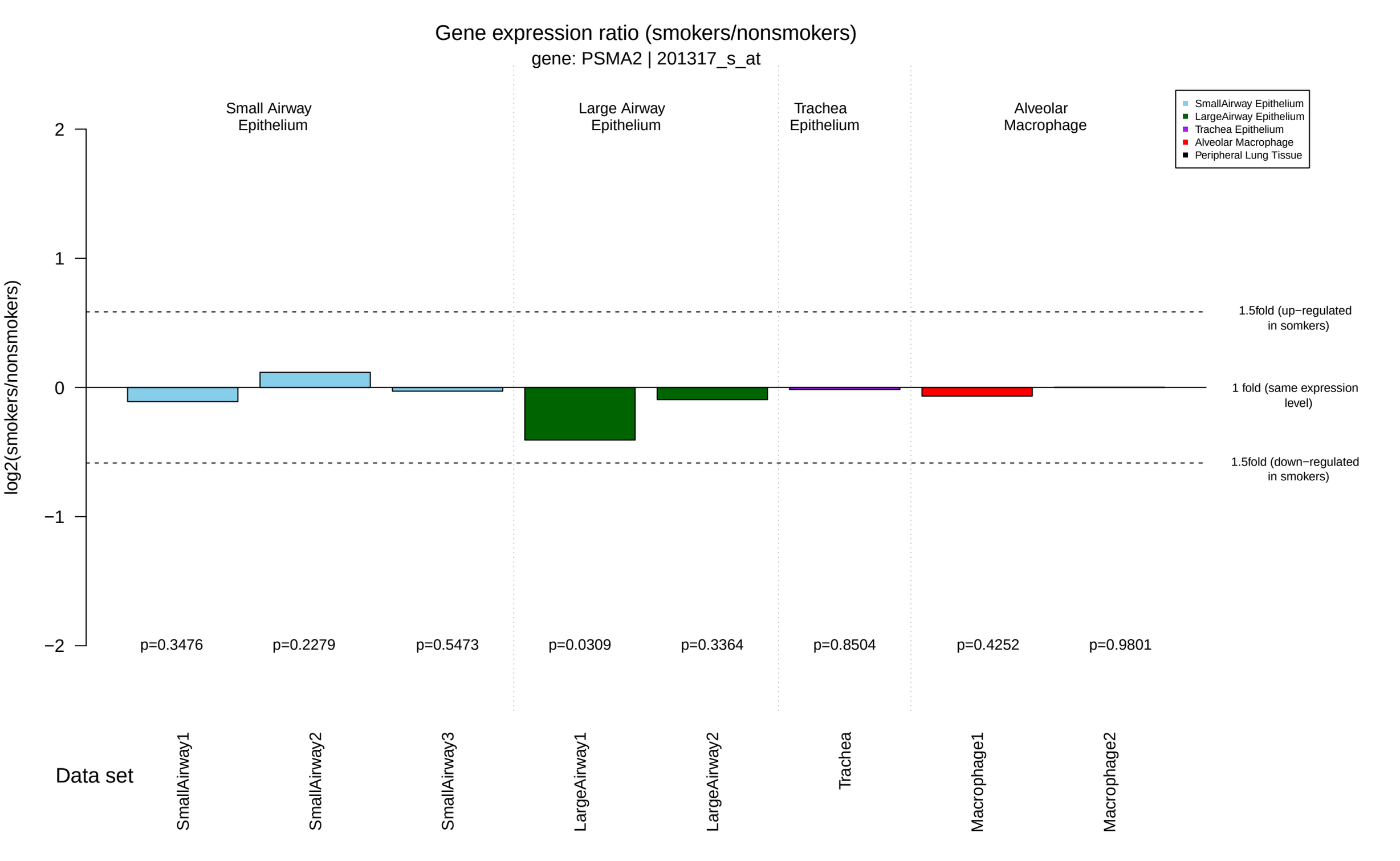
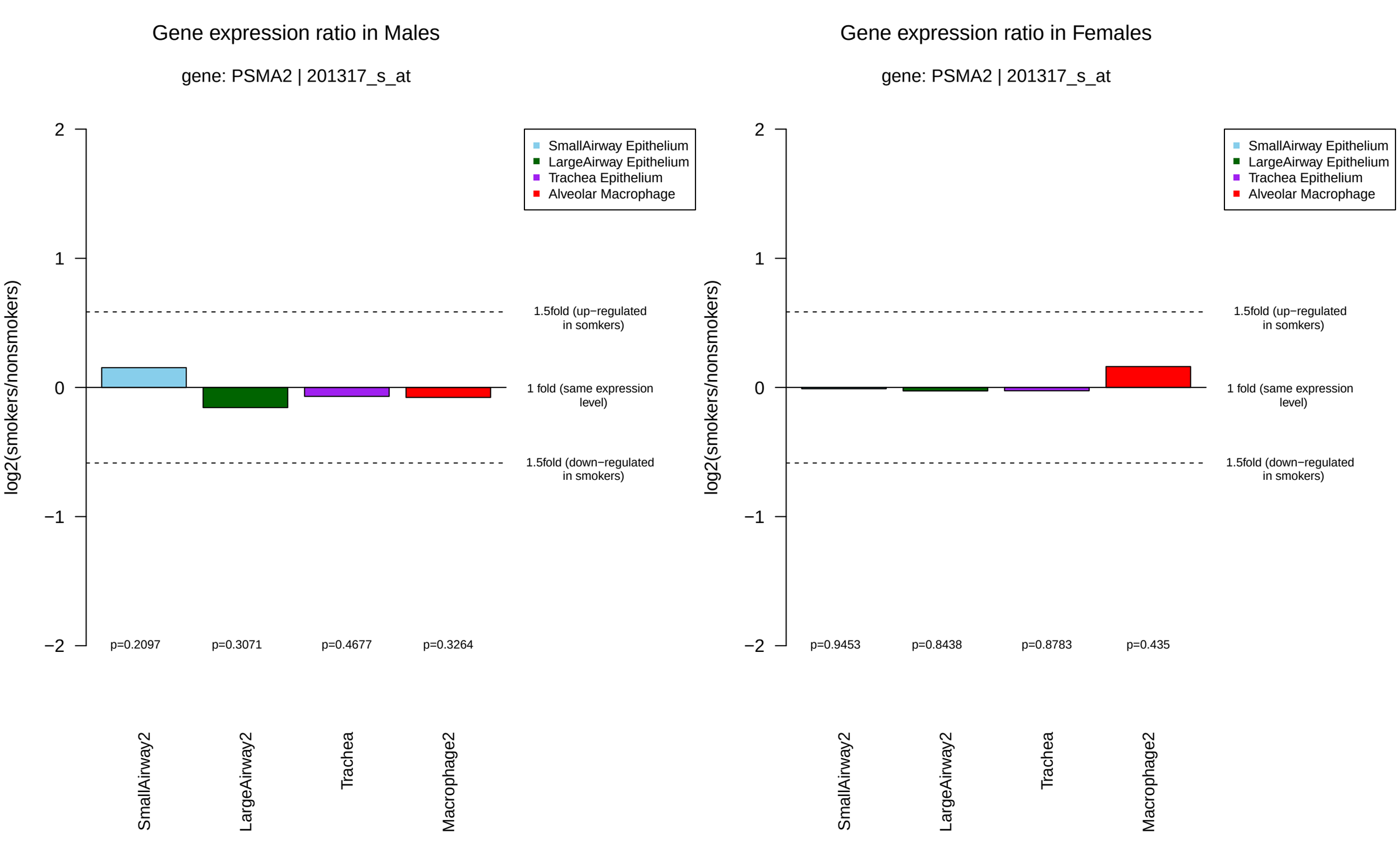
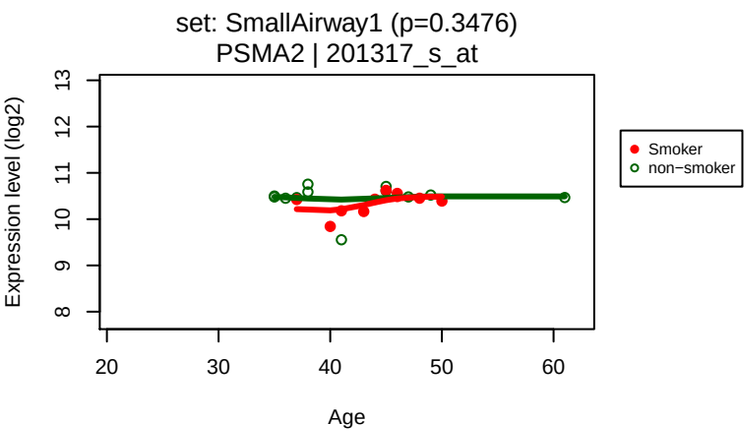
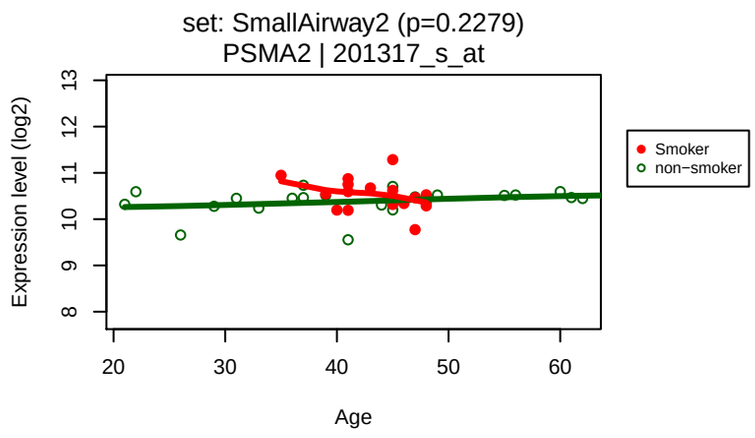
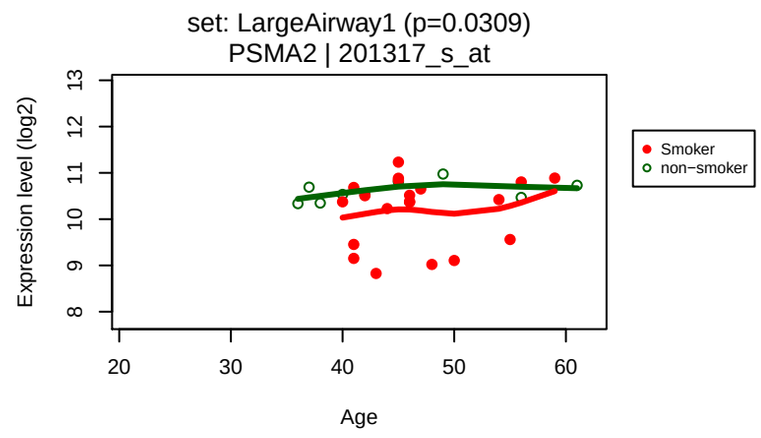

Gene expression of smokers and nonsmokers
gene: PSMA2 | 201317_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3476
p=0.2279
p=0.5473
p=0.0309
p=0.3364
p=0.8504
p=0.4252
p=0.9801
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

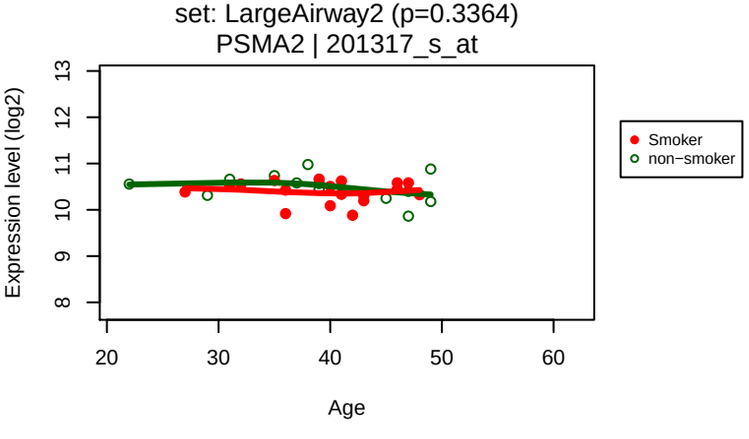

Gene expression ratio (smokers/nonsmokers)
gene: PSMA2 | 201317_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3476
p=0.2279
p=0.5473
p=0.0309
p=0.3364
p=0.8504
p=0.4252
p=0.9801
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

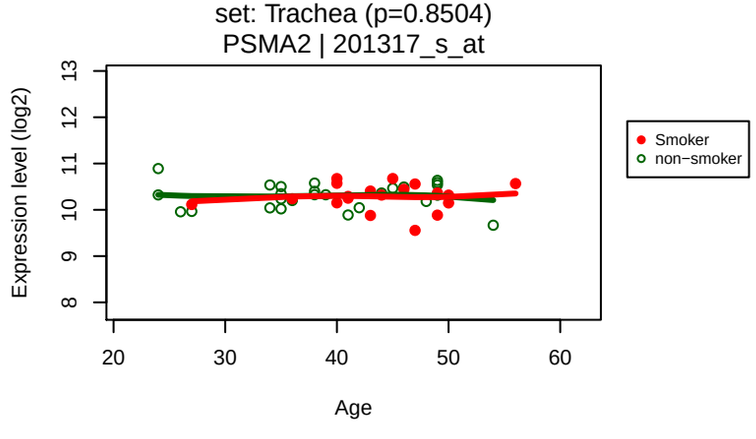

Gene expression ratio in Males
gene: PSMA2 | 201317_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2097
p=0.3071
p=0.4677
p=0.3264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA2 | 201317_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9453
p=0.8438
p=0.8783
p=0.435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

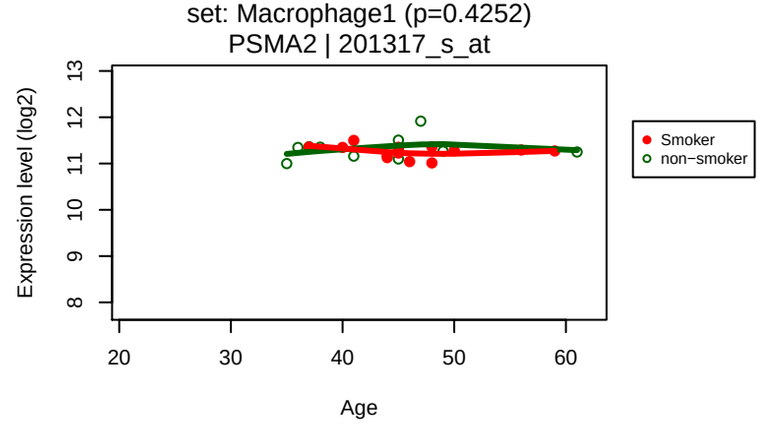

set: SmallAirway1 (p=0.3476)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker

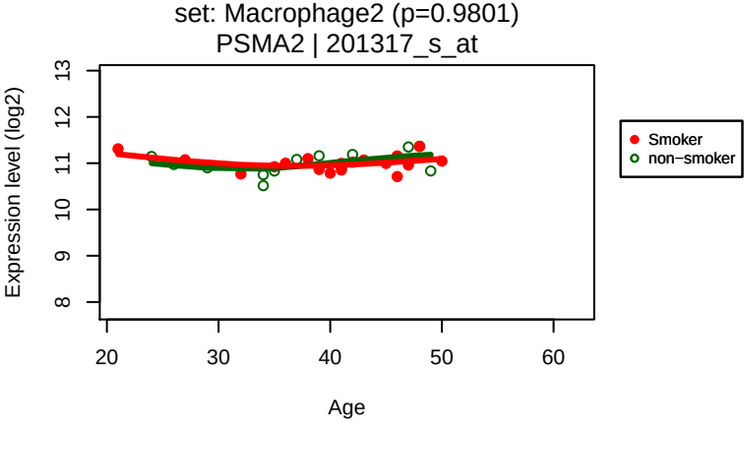

set: SmallAirway2 (p=0.2279)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker

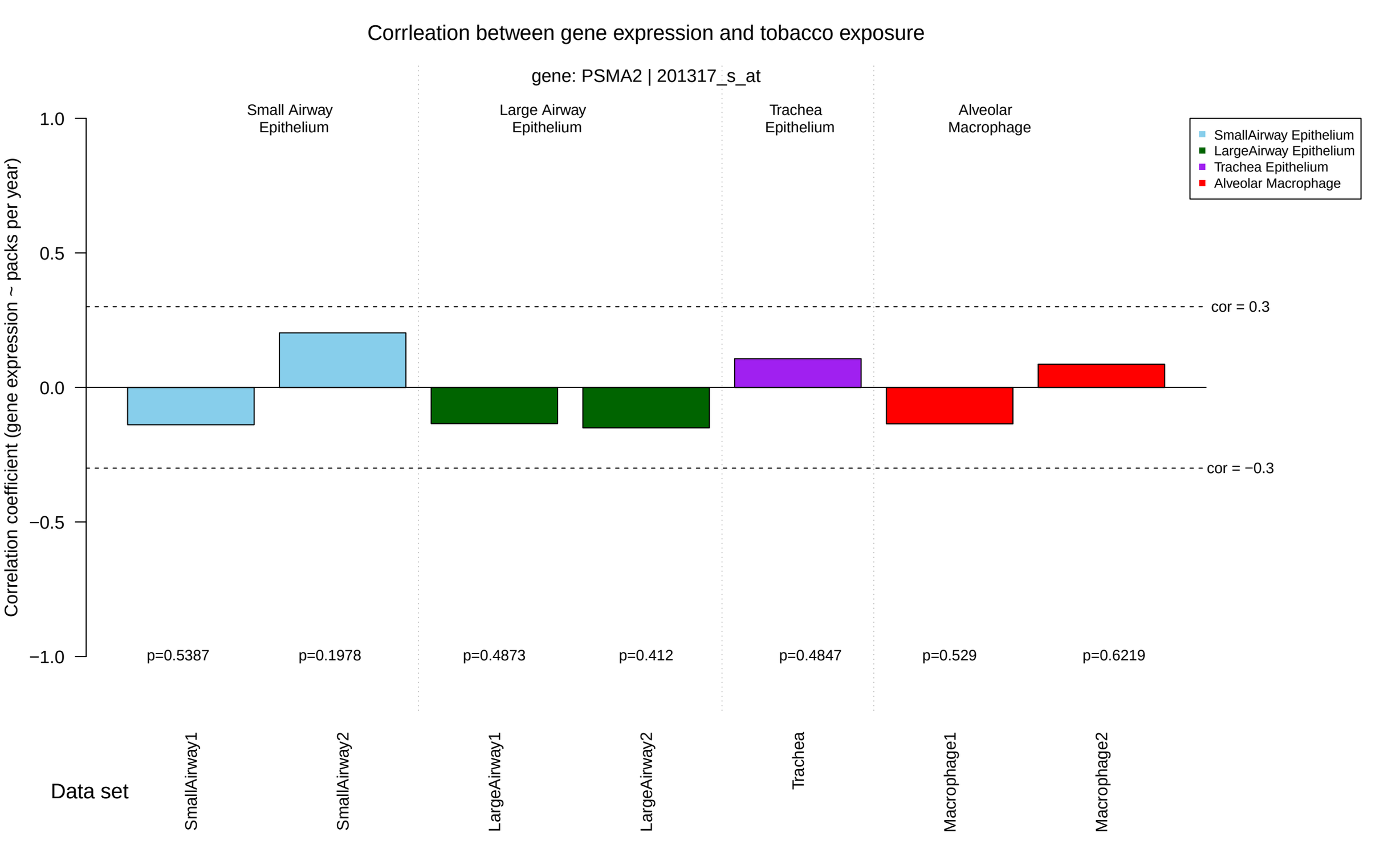

set: LargeAirway1 (p=0.0309)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3364)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8504)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4252)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9801)
PSMA2 | 201317_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA2 | 201317_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5387
p=0.1978
p=0.4873
p=0.412
p=0.4847
p=0.529
p=0.6219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2